

# Characterizing a Fiber-Based Frequency Comb With Electro-Optic Modulator

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**Abstract**—We report on the characterization of a commercial-core fiber-based frequency comb equipped with an intra-cavity free-space electro-optic modulator (EOM). We investigate the relationship between the noise of the pump diode and the laser relative intensity noise (RIN) and demonstrate the use of a low-noise current supply to substantially reduce the laser RIN. By measuring several critical transfer functions, we evaluate the potential of the EOM for comb repetition rate stabilization. We also evaluate the coupling to other relevant parameters of the comb. From these measurements, we infer the capabilities of the femtosecond laser comb to generate very-low-phase-noise microwave signals when phase-locked to a high-spectral-purity ultra-stable laser.

## I. INTRODUCTION

FEMTOSECOND lasers have revolutionized the field of time and frequency metrology by providing a phase-coherent link between a large span of optical and microwave frequencies [1]–[7]. These so-called optical frequency combs are widely used for optical frequency measurements with uncertainty close to the limitations of the cesium fountain clocks [8]–[11], as well as frequency comparisons between different optical frequency standards [12], [13]. By transferring the spectral purity of ultra-stable continuous wave (CW) lasers [14]–[18] to the microwave domain with minute excess added noise [19]–[21], they also provide ultra-low-phase-noise microwave signals [22]–[24]. Such signals may be of wide technological application in fields such as radar, telecommunications, deep-space navigation systems, timing distribution, and synchronization [25]. They have proven to be suitable as interrogation signals for atomic fountain clocks at the quantum projection noise limit [26], [27].

Among the different mode-locked laser technologies available, the titanium:sapphire (Ti:S) laser has traditionally been the workhorse of most optical metrology laboratories. Erbium-doped fiber-based systems are currently evolving into very serious contenders, especially when high reliability and very-long-term operation (several days or

weeks of operator-free continuous measurements) are crucial aspects [28]. The main drawbacks of erbium-doped fiber-based systems, when compared with Ti:S lasers, have been the higher noise and lower control bandwidth available. Laboratory systems with an intra-cavity electro-optic modulator (EOM) for improved control bandwidth have been demonstrated [29]–[32]. Highly reliable systems, including a free-space EOM actuator and exhibiting the high repetition rates suitable for optical frequency metrology, are now available commercially with a built-in self-referencing unit (Menlo Systems GmbH, Martinsried, Germany). We present in this paper several characterizations of such a commercial-core laser and demonstrate how we have used this information to understand and minimize the residual noise of the system when phase-locked to an ultra-stable cavity-stabilized CW laser, with a strong emphasis on low-phase-noise microwave signal generation. We start by presenting the various parameters and actuators of the commercial-core optical frequency comb. We proceed with a description of the methods used to measure actuators' response (transfer functions) and the relevant noise properties of our comb. We conclude by applying these studies to low-phase-noise microwave generation.

## II. OPTICAL FREQUENCY COMB

A self-referenced optical frequency comb is characterized, in the frequency domain, by a set of phase-coherent optical frequencies  $\nu_N = N \times f_{\text{rep}} + f_0$ , where  $N$  is the index of the mode,  $f_{\text{rep}}$  is the repetition rate (typically hundreds of megahertz), and  $f_0$  is the carrier-envelope offset frequency [1]. A free-running optical frequency comb exhibits fluctuations of  $f_{\text{rep}}$ ,  $f_0$ , and its average amplitude  $A$ . The state of the laser at a given time is therefore characterized by these three distinct parameters. Note that, in principle, beyond these three global parameters, one could imagine that the phase and amplitude relation between the different spectral components of the comb could also be changing over time. Because of the nonlinear effects responsible for the mode-locking mechanism, such variations have been shown to be minute and, at best, have a very slow evolution [33]. We will, therefore, not take such effects into account in this paper. A full characterization of the comb's noise properties, therefore, requires the measurement of three power spectral densities (PSD), of either phase or frequency noise for  $f_0$  and  $f_{\text{rep}}$ , and of amplitude noise for  $A$ . Note that the effect of parameter  $A$  is usually neglected in optical frequency metrology. In the

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context of low-phase-noise microwave signal generation, where amplitude-to-phase coupling in the photodetection process is a strong limitation [34], [35], characterizing the amplitude fluctuations of the laser is, however, of utmost interest.

These three parameters ( $f_0$ ,  $f_{\text{rep}}$ , and  $A$ ), can be controlled via multiple actuators. In this paper, we characterize a femtosecond comb system which possesses a piezo-electric actuator (PZT, whose purpose is to change the laser cavity length), allows fine tuning of the laser pumping via adjustment of the current which drives the pump laser diodes, and has an EOM in the cavity. The latter element operates mainly as a very fast voltage-controlled group delay element in the femtosecond laser's cavity. It therefore changes the repetition rate, although with unavoidable coupling to the other two parameters ( $f_0$  and  $A$ ). Ideally, the PZT modifies only  $f_{\text{rep}}$ . However, it is well known that a tiny misalignment of the laser cavity may influence both  $f_0$  and  $A$ . Changing the pump power of the laser comb (via the current driving the pump diode lasers) obviously impacts  $A$ , as well as  $f_0$  and  $f_{\text{rep}}$  via complex mode-locked laser dynamics [36]–[38]. A full characterization of the response of the comb to the various actuators therefore requires the measurement of the nine transfer function matrix elements (three actuators for three comb parameters) as shown by

$$\begin{pmatrix} A(j\omega) \\ f_{\text{rep}}(j\omega) \\ f_0(j\omega) \end{pmatrix} = \begin{bmatrix} H_{I_{\text{pump}}, A}(j\omega) & H_{V_{\text{PZT}}, A}(j\omega) & H_{V_{\text{EOM}}, A}(j\omega) \\ H_{I_{\text{pump}}, f_{\text{rep}}}(j\omega) & H_{V_{\text{PZT}}, f_{\text{rep}}}(j\omega) & H_{V_{\text{EOM}}, f_{\text{rep}}}(j\omega) \\ H_{I_{\text{pump}}, f_0}(j\omega) & H_{V_{\text{PZT}}, f_0}(j\omega) & H_{V_{\text{EOM}}, f_0}(j\omega) \end{bmatrix} \begin{pmatrix} I_{\text{pump}}(j\omega) \\ V_{\text{PZT}}(j\omega) \\ V_{\text{EOM}}(j\omega) \end{pmatrix}, \quad (1)$$

where  $I_{\text{pump}}$  is the pump diode current,  $V_{\text{PZT}}$  is the voltage driving the PZT, and  $V_{\text{EOM}}$  is the voltage driving the EOM crystal. We have measured the nine transfer functions of this matrix, but we will focus on the subset which impact optical frequency metrology and the optical-to-microwave division process.

We proceed with a description of the methods we developed to measure these transfer functions, as well as several noise properties of the femtosecond laser.

### III. TRANSFER FUNCTIONS AND NOISE MEASUREMENTS TECHNIQUES

Our commercial-core fiber-based frequency comb has a repetition rate near  $f_{\text{rep}} \sim 250$  MHz, which can be coarsely adjusted to within  $\pm 1$  MHz using a motorized translation stage on one of the cavity mirrors. The available optical output power is about 130 mW. The laser has a built-in  $f$ - $2f$  interferometer unit which generates the offset frequency signal  $f_0$ . A motorized wedge [38] allows coarse frequency adjustment of this quantity, which we maintain near 70 MHz. To measure transfer functions, we use a

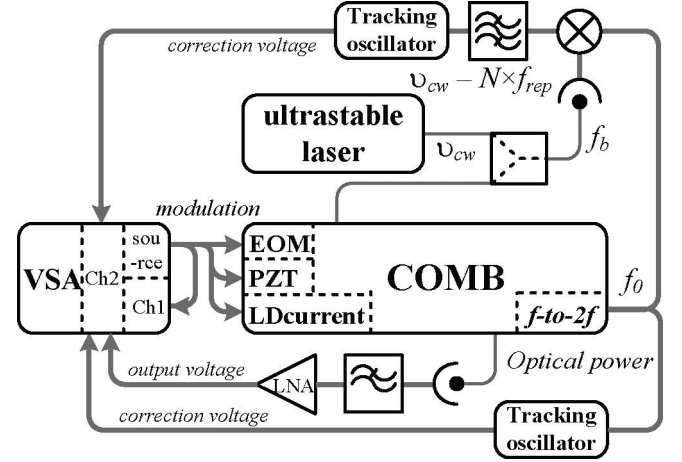


Fig. 1. Setup schematic used to measure transfer functions of amplitude,  $f_0$ , and  $f_{\text{rep}}$  versus laser pump current, intracavity EOM voltage, and PZT voltage, as well as power spectral densities of the same quantities.  $f$ -to- $2f$ : built-in self-referencing unit; LNA: low-noise amplifier; VSA: vector signal analyzer.

two-channel vector signal analyzer (VSA; 89410A, Agilent Technologies, Santa Clara, CA) which can operate at up to 10 MHz Fourier frequency and has a built-in programmable source voltage output. The applied modulation is directed simultaneously to the channel 1 input of the VSA and the actuators of the laser that we want to characterize (PZT, pump diode current, or EOM) via a suitable actuator driver. Note that these drivers must be independently characterized, to ensure they don't impose bandwidth limitation to the measurement. The quantity to characterize ( $f_{\text{rep}}$ ,  $f_0$ , or  $A$ ) is transformed into a voltage signal and fed to channel 2 of the VSA. By programming the VSA source output to generate a chirped sine wave, we measure the transfer function from channel 1 to channel 2. This allows the transfer function from the actuator to the comb parameter to be obtained. The set-up schematic is depicted in Fig. 1.

The key part of such an experiment is to design ways to transform  $f_{\text{rep}}$ ,  $f_0$ , and  $A$  into a voltage signal with high sensitivity and minimal crosstalk from the other two parameters of the comb. Measuring  $A$  is quite straightforward. We inject a few milliwatts of light from the femtosecond laser output (30-nm bandwidth around 1.55  $\mu\text{m}$ ) onto a fiber-pigtailed InGaAs photodiode. The output of the photodiode is low-pass filtered to remove the harmonics of the repetition rate and, after low-noise amplification, the output voltage is fed to the VSA. To measure  $f_0$  fluctuations, we process this signal with a home-made fast tracking oscillator filter. This device is composed of a voltage-controlled oscillator (VCO), which is phase-locked to the input signal with more than 3 MHz of bandwidth. At Fourier frequencies lower than this bandwidth, the correction signal of the tracking oscillator (i.e., the voltage which controls the VCO when the phase-locked loop is operating) is proportional to the frequency fluctuations of  $f_0$  (the free-running VCO noise is negligible compared with that of  $f_0$ ). To measure  $f_{\text{rep}}$  fluctuations, it is convenient

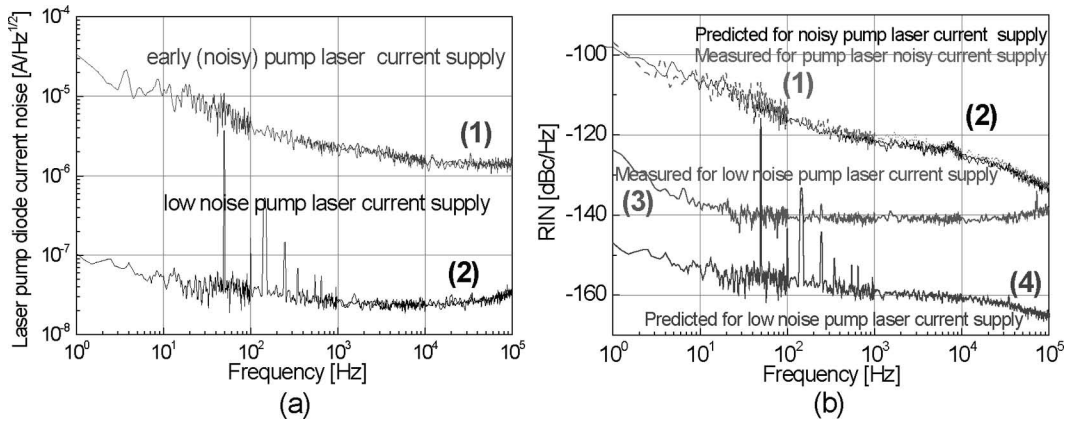


Fig. 2. (a) Measured spectral density of current noise for early noisy [curve (1)] and low-noise [curve (2)] current supplies. (b) Relative intensity noise (RIN) of the femtosecond laser output with two different current supplies for the pump lasers diodes. Curves (1), measured, and (2), predicted, are for the early (noisy) current supply. Curves (3), measured, and (4), predicted, are for the low-noise current supply.

to increase the sensitivity by measuring  $N \times f_{\text{rep}}$ , where  $N$  is a large integer. By beating the comb output with an ultra-stable laser (linewidth  $< 1$  Hz) [17] of optical frequency  $\nu_{\text{cw}}$  (near the 1.55- $\mu\text{m}$  central wavelength of the comb), we have access to  $f_b = N \times f_{\text{rep}} + f_0 - \nu_{\text{cw}}$ . This RF signal, mixed with  $f_0$ , leads to two sidebands that are highly sensitive to  $f_{\text{rep}}$  fluctuations (because of the large multiplicative factor  $N$ ). We select, with a band-pass filter, the sideband which is independent from  $f_0$  (because  $f_b - f_0 = N \times f_{\text{rep}} - \nu_{\text{cw}}$ ). Locking a tracking oscillator to this signal, therefore, leads to a voltage-correction signal proportional to the fluctuations of  $f_{\text{rep}}$ .

We use these techniques to transform  $A$ ,  $f_0$ , and  $f_{\text{rep}}$  into voltage signals to measure the nine transfer functions and the noise power spectral densities of the optical frequency comb.

#### IV. RESULTS AND APPLICATIONS

Once the comb has been characterized by the  $3 \times 3$  transfer functions matrix, we can use this information to improve the comb's properties. Our work puts a strong focus on the context of low-phase-noise microwave signal generation by photodetection of the repetition rate's harmonics when the comb is phase-locked to an ultra-stable CW laser [23], [24], [27], [39]. However, the use of the transfer functions matrix goes well beyond what appears in this paper. A first example of the use of the transfer function/noise characterization is illustrated in Fig. 2.

The first comb was equipped with a relatively noisy pump diode laser current supply [Fig. 2(a), curve (1)]. We first measure the pump diode current-to-comb amplitude transfer function which can be simply modeled by a first-order low-pass filter (roll-off  $\sim 60$  kHz) with a coefficient of  $\sim 4 \times 10^{-3}$  W/A. By plugging the current noise of this current supply into the transfer function we measured from the pump diodes' current to the amplitude of the laser, we obtain a predicted relative intensity noise (RIN) of the comb which matches the actual measurement

very well [Fig. 2(b), curves (1) and (2)]. We can conclude from this that the rather high noise of the early prototype current supply is the origin of the excessive RIN of the laser comb. Indeed, by using a low-noise current supply as a replacement [Fig. 2(a), curve (2)], we obtained a very substantial improvement of the RIN. Furthermore, by comparing the new RIN measurement [Fig. 2(b), curve (3)] and the predicted RIN [Fig. 2(b), curve (4)] for the low-noise current supply, we can conclude that the current supply is no longer a limiting factor.

By studying the RIN of the pump laser diode itself (when driven by the low-noise current supply), we observed a RIN higher than expected from the current supply's characteristic alone. This excess noise explains the comb's RIN for Fourier frequencies lower than 30 to 50 kHz. Improving the performance may require hand-selecting lower-RIN pump laser diodes and/or actively controlling their output power. For Fourier frequencies higher than 30 to 50 kHz, the RIN of the pump diode lasers does not explain the RIN of the comb. Understanding this excess noise at high Fourier frequencies would necessitate further theoretical and experimental studies.

A second example of transfer function application is to optimize the feedback loop when phase-locking the comb to an external reference (an ultra-stable 1.542- $\mu\text{m}$  laser in our case). In Fig. 3, we present the amplitude and phase of the transfer function from the EOM control voltage to  $f_{\text{rep}}$ . From this data, we deduce that the maximum available servo bandwidth when using the EOM as an actuator to control  $f_{\text{rep}}$  is limited by sharp resonances observed around 900 kHz. As a matter of fact, they produce rapid large phase shifts so large that they cannot be easily compensated by control circuit design. We use proportional and multiple integration control which acts on both the EOM and the PZT actuators (both act mainly on  $f_{\text{rep}}$ ) to phase-lock  $f_{\text{rep}}$  to a CW ultra-stable laser near 1.55  $\mu\text{m}$ . Similarly to the previously described technique for measuring  $f_{\text{rep}}$  fluctuations, the beat note between the CW laser and one nearby tooth of the comb is mixed with  $f_0$  to provide a signal of frequency  $N \times f_{\text{rep}} - \nu_{\text{cw}}$ , independent of  $f_0$ .

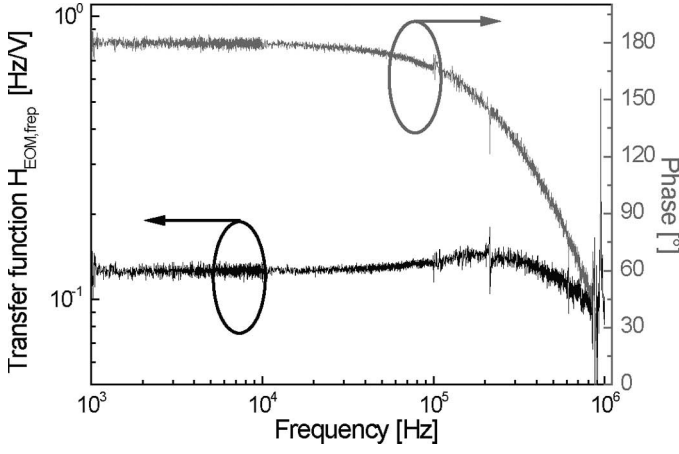


Fig. 3. Transfer function  $H_{V_{EOM}f_{rep}}$  from electro-optic modulator (EOM) voltage to repetition rate  $f_{rep}$  in amplitude (bottom curve, left axis) and phase (top curve, right axis). The ultimate bandwidth available with such an actuator is limited by the strong narrow resonances near 900 kHz Fourier frequency.

By mixing it with a reference from a fixed synthesizer, we produce an error signal which is sent to a proportional-integral (PI) controller. This PI controller directly steers the EOM. The EOM control voltage is further integrated and fed to the PZT. In this way,  $f_{rep}$  is phase-locked to the optical frequency  $\nu_{cw}$ , with the EOM acting for Fourier frequencies larger than 6 to 10 kHz, whereas the PZT controls the laser for low Fourier frequencies. Fig. 4(a) shows the in-loop error spectral density and Fig. 4(b) the correction voltage spectral density applied to the EOM when the phase-locked loop is running.

The in-loop error exhibits a resonance around 900 kHz and a servo bump near 400 kHz, consistent with the transfer function presented in Fig. 3. By integrating the in-loop error up to 1 MHz, the total phase error is about 0.21 rad rms, resulting in 96.7% energy in the carrier. Because the total phase error is much lower than 1 rad rms, the comb is in the so-called ultra-stable regime, where each tooth of the comb has nearly the same spectral purity as the ultra-stable CW laser used as a reference for the lock (once  $f_0$  is removed) [31], [40], [41].

This was confirmed independently by beating the comb output near 1062.5 nm (obtained from the same highly nonlinear fiber used in the built-in  $f-2f$  unit) with a second ultra-stable laser [42], [43]. Fig. 5 shows both the Allan deviation and the RF spectrum of the beat-note signal. The measured linewidth of the beat-note signal and the frequency stability are comparable with independently measured lasers' frequency stability. These support the statement that the additional noise introduced by the femtosecond laser comb is below the laser's frequency noise/stability. The integrated correction voltage is 1.7 Vrms ( $\sim 6$  Vpp maximum, verified with an oscilloscope). When comparing to the 25-V output range of our fast control electronics, this leaves room to optimize the EOM dimensions, using shorter crystals with lower dispersion and, po-

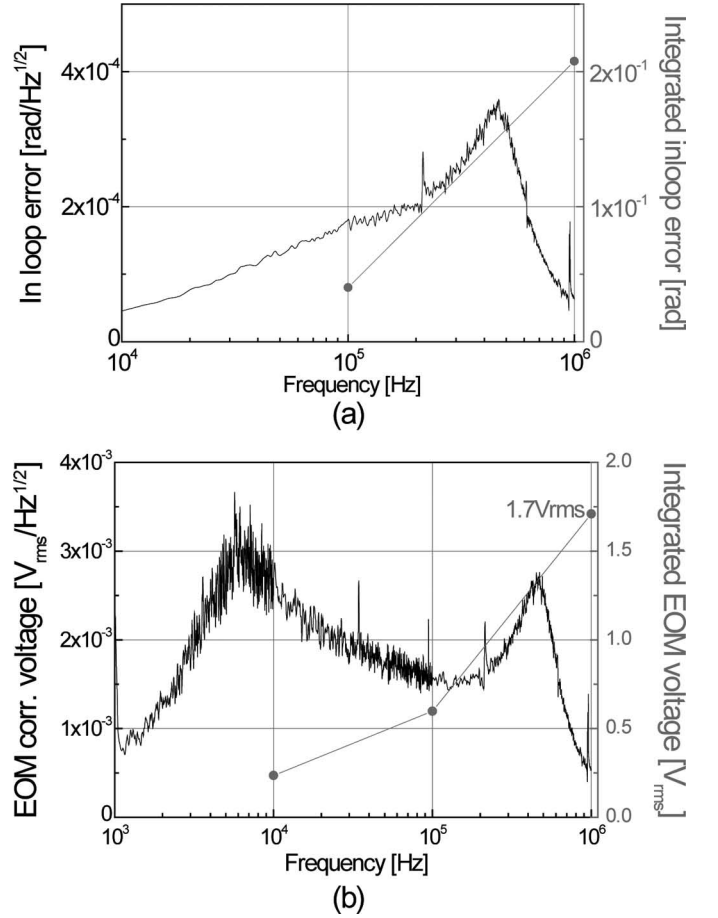


Fig. 4. (a) In-loop residual phase error signal when phase-locking the comb to a continuous ultra-stable laser near 1.55  $\mu\text{m}$ . The integrated phase error signal (right axis) is shown in gray. (b) Root mean square correction voltage applied to the electro-optic modulator (EOM) when the phase-locked loop is running. The integrated rms voltage applied to the EOM (right axis) is shown in gray. The decrease of the correction voltage below 6 kHz is due to the use of the piezoelectric actuator controlling the laser for low Fourier frequencies (in lieu of the EOM).

tentially, higher resonances which may allow larger control bandwidths.

A third example of the use of the transfer function, presented in Fig. 6, is related to the off diagonal elements of the transfer matrix (1). This figure represents the unavoidable crosstalk of the EOM control voltage to  $f_0$  and  $A$ . (Ideally, only  $f_{rep}$  would be affected.) Understanding the dynamics of these two crosstalk responses (enhanced responses at high Fourier frequencies in particular) is not straight-forward. We believe them to be caused by minute misalignments of the EOM crystal, which couples, via complex laser dynamics, to the polarization and amplitude of the pulse in the femtosecond laser's cavity. Verifying this hypothesis would require a thorough theoretical and experimental analysis which is well beyond the scope of this paper. We emphasize that the difference in dynamic response of  $f_{rep}$  compared with that of  $A$  and  $f_0$  is an important phenomenon which should be accounted for when trying to extend the classic rubber-band model



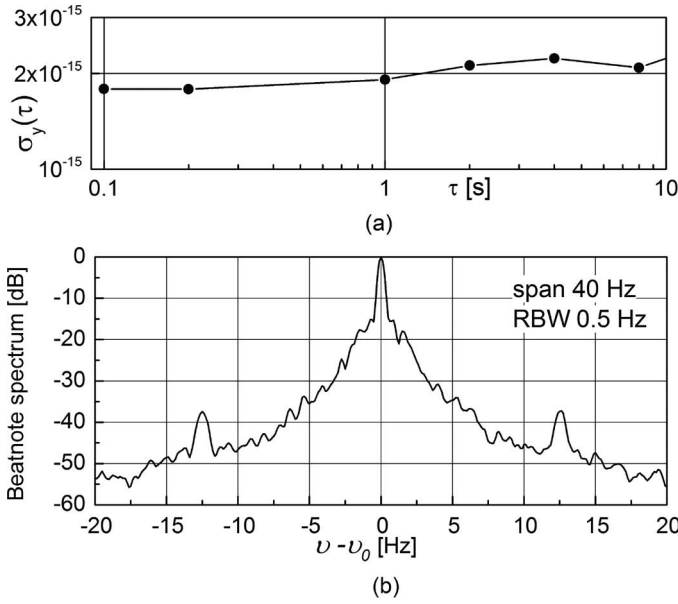


Fig. 5. (a) Allan deviation  $\sigma_y(\tau)$  of the beat-note signal between the stabilized comb and an ultra-stable laser near 1062.5 nm. (b) The radio frequency spectrum of the beat-note signal [span 40 Hz, resolution bandwidth (RBW) 0.5 Hz, 25 averages].

(introduced in the seminal paper by Telle *et al.* [44]) to high Fourier frequencies. The measured transfer functions of Fig. 6 provide some useful information about the limit of the EOM (in its current implementation) as an actuator to phase-lock  $f_{\text{rep}}$ .

By combining the correction voltage spectral density applied to the EOM when the servo loop is operating [Fig. 4(b)] with the transfer functions of Fig. 6, we can deduce the EOM control-induced fluctuations of  $f_0$  and  $A$ . The fluctuations of  $f_0$  are not important in the context of our scheme, where  $f_0$  is uncontrolled and simply removed from  $f_b$  by frequency mixing. Other  $f_0$ -removal techniques, including the one presented in [45], are likewise inherently immune to such crosstalk. However, note that experiments which independently phase-lock  $f_{\text{rep}}$  and  $f_0$ , as in [8], [46], and [47], may need to account for those effects. On the other hand, the crosstalk-induced fluctuations of  $A$  are of pre-eminent importance in the context of low-noise microwave signal extraction by photodetection of the femtosecond pulses. Indeed, amplitude-to-phase conversion which occurs in the photodetection process limits the achievable performance of low-phase-noise microwave generation experiments [34], [35]. From the predicted crosstalk-induced fluctuations of  $A$ , and a given amplitude-to-phase conversion factor related to the photodetection process, one can deduce the achievable level of residual microwave phase noise. For instance, in our case, for a microwave signal at 12 GHz generated from a 250-MHz repetition rate laser, we assume a  $d\phi/(dP/P) \sim 0.03$  rad amplitude-to-phase conversion factor of the photodetection process. Here,  $\phi$  is the phase of the microwave compared with that of the pulse train and  $P$  is the average optical power on the photodetector. Note that substantially higher conversion

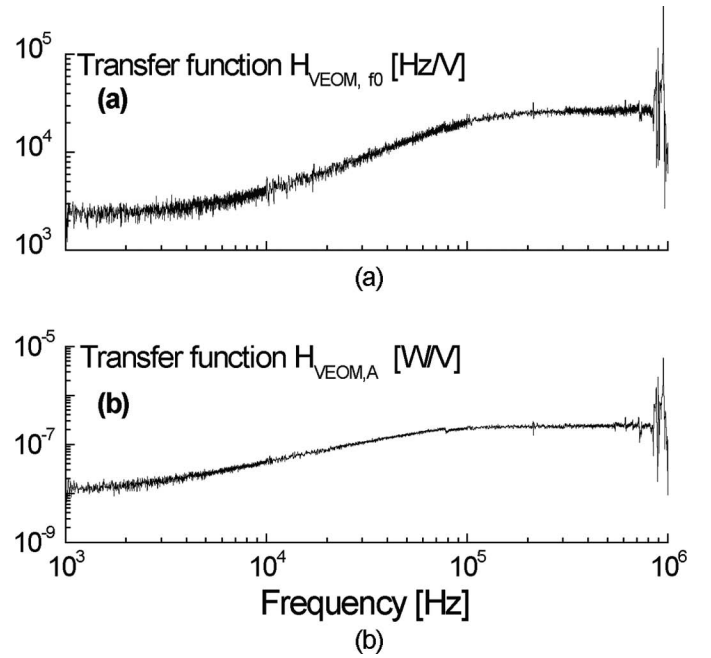


Fig. 6. (a) Transfer function  $H_{V_{\text{EOM}}, f_0}$  [electro-optic modulator (EOM) voltage to  $f_0$ ]. (b)  $H_{V_{\text{EOM}}, A_0}$  (EOM voltage to comb's amplitude).

factors are normally obtained without the fine adjustment of the experimental parameters [34], [35].

Fig. 7 presents the predicted residual phase noise limits obtained from RIN-induced phase noise [curve (c)] and EOM crosstalk-induced phase noise [curve (d)] value of the amplitude-phase conversion factor. When comparing these predictions to the best noise floor limit we measured for such a system with a repetition rate multiplication technique detailed in [48], it appears that the crosstalk is not a limiting factor at the present level of noise floor limit. Note, however, that if the amplitude-phase conversion factor was equal to 1 rad (typical worst-case scenario observed from [39]), it would constitute a limitation for Fourier frequencies in the 10 kHz to 1 MHz range that would be substantially higher than the measured noise floor limit. Furthermore, the RIN-induced residual phase noise is also lower than the noise floor limit. Similarly, an amplitude-phase conversion factor higher than 0.1 rad would induce a substantial limitation to the performances of the system. On the other hand, it also appears from Fig. 7 that the main limitation in the 100 kHz to 1 MHz Fourier frequencies range is due to the residual in-loop error of the phase-locked loop [curve (b)]. This limitation is ultimately linked to the still-limited actuation bandwidth achievable with the EOM. Further improvement of the EOM actuator would therefore be necessary to achieve residual phase noise limits in the  $-160$  dB(rad<sup>2</sup>/Hz). Note that a 1.2-MHz bandwidth was achieved with a waveguide EOM [30] and we could therefore expect a similar result for future prototypes. Since submission of this work, we have tested a second unit showing smooth piezoelectric resonances beyond 2 MHz and we have demonstrated locking bandwidths larger than 1.2 MHz.

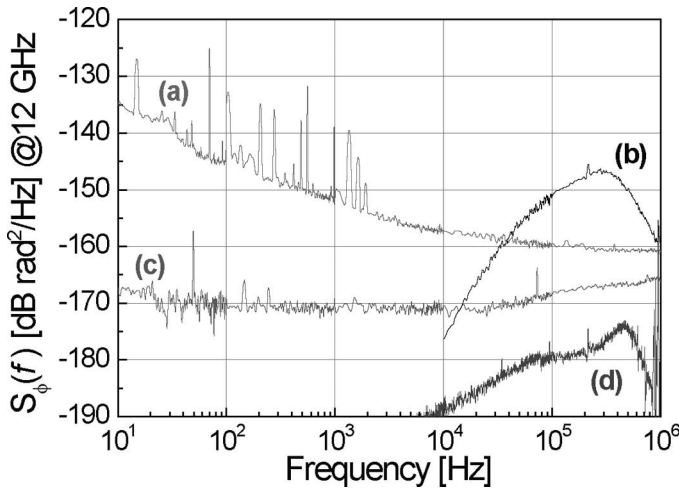


Fig. 7. Residual phase noise for optical-to-microwave frequency division operation of the frequency comb at 12 GHz. Curve (a): detection noise floor limit (shot noise at high Fourier frequencies) from [43]; curve (b): phase noise induced by electro-optic modulator (EOM) in-loop error (i.e., resulting from limited bandwidth of the EOM actuator); curve (c): relative intensity noise (RIN)-induced phase noise; curve (d): EOM voltage to amplitude-induced phase noise. To obtain curves (c) and (d), we assumed typical 0.03 rad per relative amplitude change of the amplitude-to-phase conversion factor in the photodetection process.

## V. CONCLUSIONS

We have presented in detail the techniques used to measure the various transfer functions which characterize an optical frequency comb equipped with various actuators. We have shown some examples of how to use such characterization to deduce the performance limits of the comb and improve its performance. When phase-locking the repetition rate of the comb to an ultra-stable CW laser optical frequency, we reached the ultra-stable regime, in which the linewidth of each comb tooth is limited by that of the CW laser reference (and not the phase-locked loop performances). In the context of low-phase-noise microwave generation by photodetecting the pulse train, we have made a prediction of the achievable minimum phase noise from different possible noise sources. These predictions make us confident that we can achieve residual phase noise in the  $-160$  dB(rad<sup>2</sup>/Hz) level, close to the shot noise limit, with a slightly improved EOM actuator. In addition, we have shown how to use the dynamic characterization of a frequency comb (via its actuators' transfer functions) to understand and improve an optical frequency comb's performance.

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Authors' photographs and biographies were unavailable at time of publication.